TEAM: SLU 0501 DVT TEAM 22

UNLOCKING INSIGHTS: TURNING DATA INTO BUSINESS DECISIONS

Team Members: Abu Yussif, Aishat Adeniyi, Ayesha Ashraf, Mpfariseni Ephraim

Tools Used: Python (Pandas, Matplotlib), Google Looker Studio, Microsoft Excel, Power BI

EXECUTIVE SUMMARY

This report presents the outcome of a **three-week** data analytics project focused on optimizing DePaul University's international student outreach efforts.

Our team transformed a raw dataset of over 7,500 records into an interactive strategic dashboard aimed at improving enrollment outcomes and outreach efficiency.

INTRODUCTION

The **purpose** of this analysis was to explore DePaul University's outreach and application data to identify trends, performance gaps, and optimization opportunities.

Context: Global diversity remains a key strategic priority for DePaul University .

Challenges Identified:

- Data noise (80 columns with inconsistent entries)
- Limited funnel visibility
- Measuring counselor ROI

Objective: Transform raw outreach data into actionable insights for data-driven recruitment strategy.

METHODOLOGY

Data Understanding (EDA): Explored data to map current outreach state.
Data Cleansing: Utilized Python scripts to normalize dates and categorical values.
Feature Engineering: Created new variables such as Is_Successful to measure performance.
Data Visualization: Built interactive dashboards using Looker Studio and Power BI.

Framework: Data → Insight → Action

```
df['Date_of_Contact'] = pd.to_datetime(df['Date_of_Contact'])
df['Admit_Date'] = pd.to_datetime(df['Admit_Date'])
df['Recieved_At-2'] = pd.to_datetime(df['Recieved_At-2'])
```

```
df.drop_duplicates('Reference_ID',inplace=True)
```

```
df.info()
```

DATASET OVERVIEW

Source: Excel export containing **7,543** student records and 80 columns.

Audit findings:

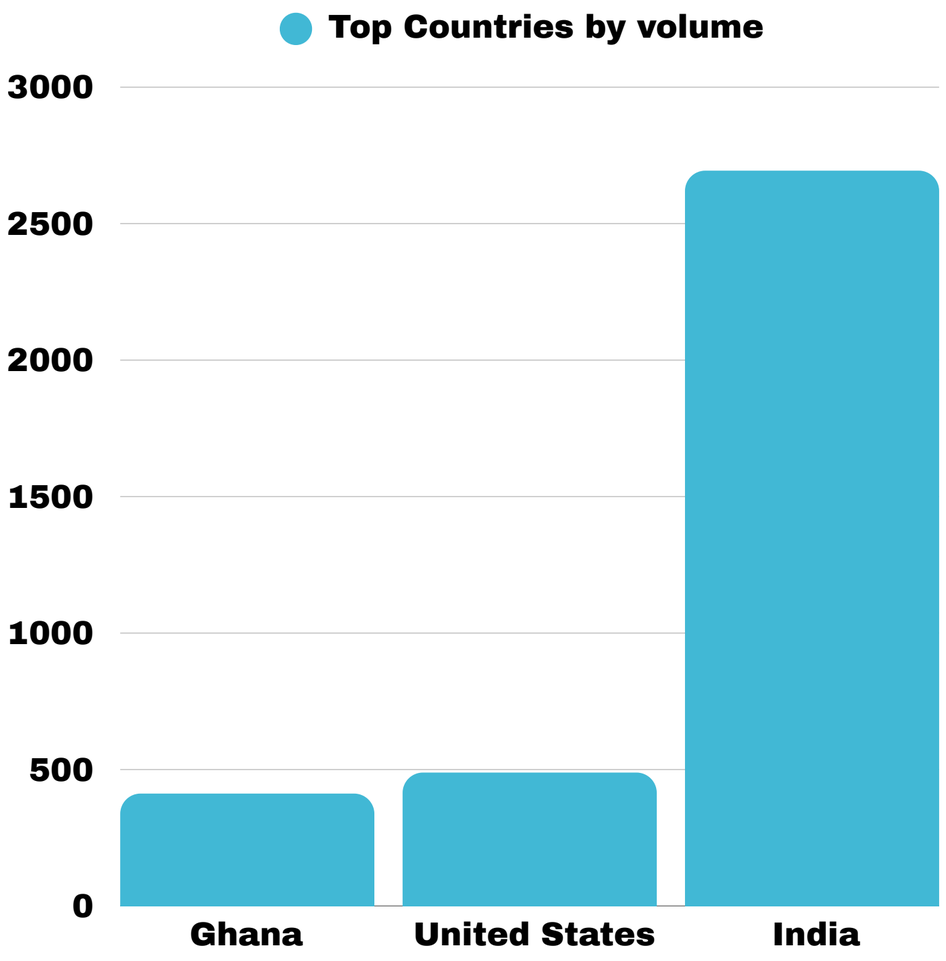
- **38** columns were **100%** null.
- **81%** columns contained missing values.
- **5,000** unique student entries identified.

GOAL

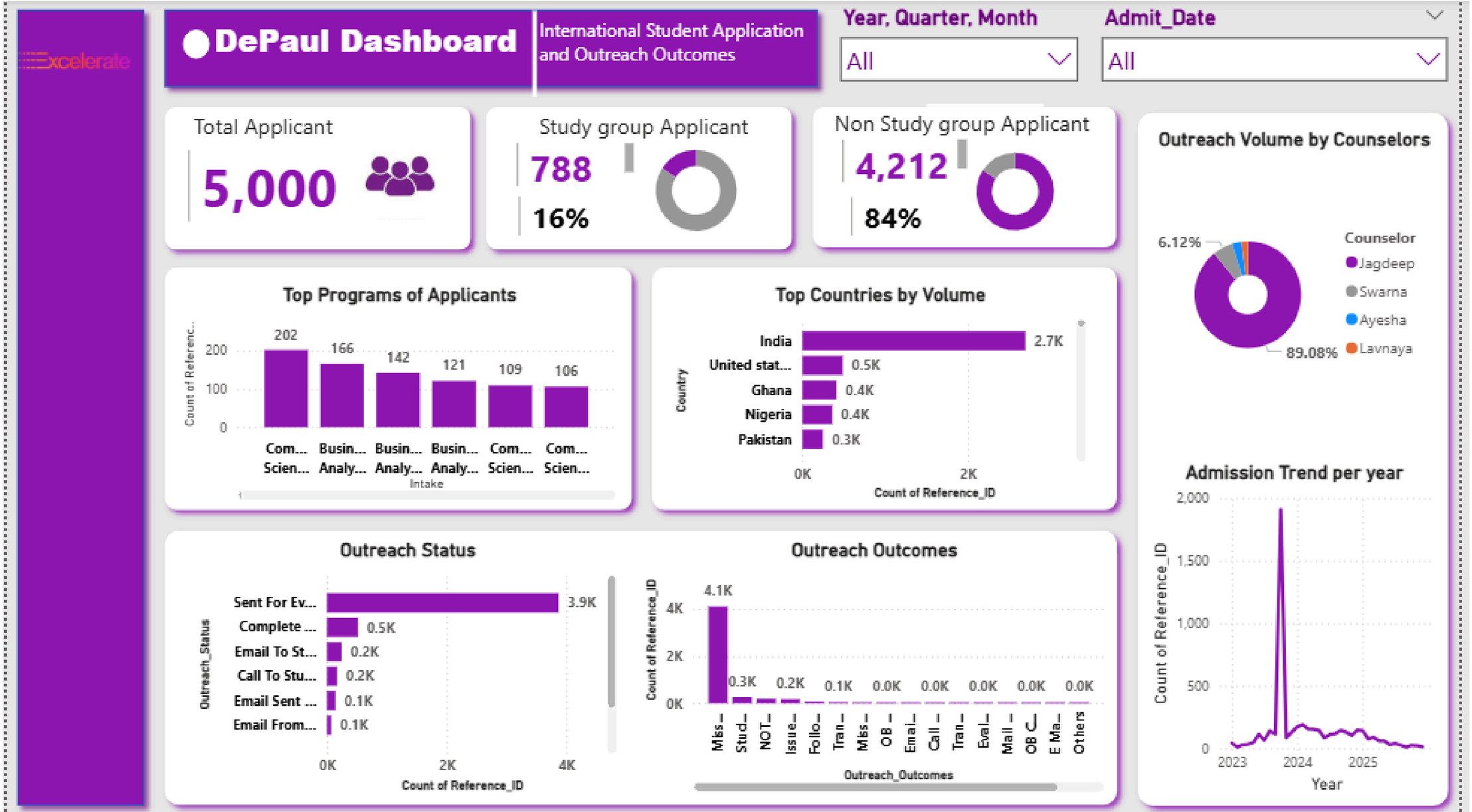
Simplify and **standardize** the dataset for **accurate** and **efficient** analysis.

KEY PERFORMANCE INDICATORS (KPIS)

- **5,000** unique applicants identified.
- Applicants from Study Group: **788 (16%)**; Non-Study Group: **4,212 (84%)**.
- **Outreach effectiveness**: **82%** of applicants had no missing transcripts
- **Top-performing counselor:** Japdeep (handled **89%** of leads).
- **Geographic concentration: 53%** of leads originated from India.
- **Outreach efficiency: 40%** inactive leads — major lost opportunities.

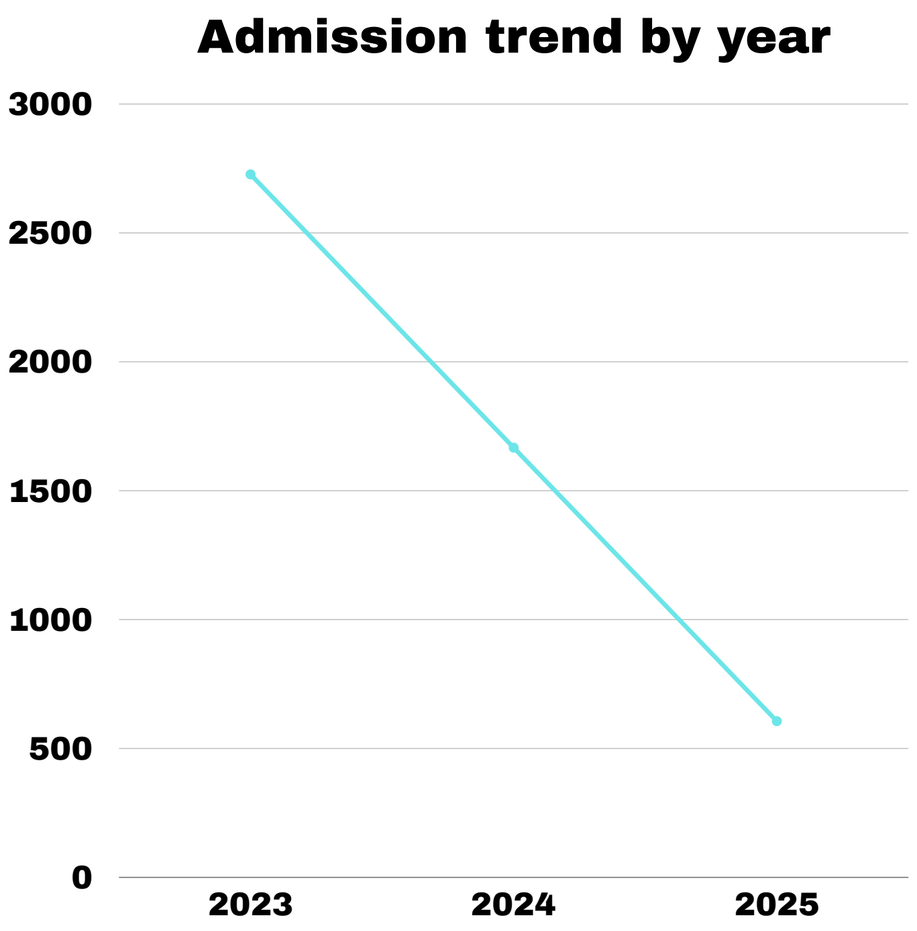


DASHBOARD


Excelerate
DePaul Dashboard
International Student Application and Outreach Outcomes
Year, Quarter, Month
All
Admit_Date
All
Total Applicant
5,000
Study group Applicant
788
16%
Non Study group Applicant
4,212
84%
Outreach Volume by Counselors
6.12%
89.08%
Counselor
Jagdeep
Swarna
Ayesha
Lavnaya
Top Programs of Applicants
Count of Referenc...
202
166
142
121
109
106
Com... Scien...
Busin... Analy...
Busin... Analy...
Busin... Analy...
Com... Scien...
Com... Scien...
Intake
Top Countries by Volume
Country
India 2.7K
United stat... 0.5K
Ghana 0.4K
Nigeria 0.4K
Pakistan 0.3K
0K
2K
Count of Reference_ID
Admission Trend per year
Count of Reference_ID
2,000
1,500
1,000
500
0
2023
2024
2025
Year
Outreach Status
Outreach_Status
Sent For Ev... 3.9K
Complete ... 0.5K
Email To St... 0.2K
Call To Stu... 0.2K
Email Sent ... 0.1K
Email From... 0.1K
0K
2K
4K
Count of Reference_ID
Outreach Outcomes
Count of Reference_ID
4K
2K
0K
4.1K
0.3K
0.2K
0.1K
0.0K
0.0K
0.0K
0.0K
0.0K
Miss...
Stud...
NOT...
Issue...
Follo...
Tran...
Miss...
OB ...
Emai...
Call ...
Tran...
Eval...
Mail ...
OB C...
E Ma...
Others
Outreach_Outcomes

KEY INSIGHTS

- Counselor performance gap revealed significant inconsistency.
- Outreach volume ≠ effectiveness — quantity didn’t guarantee results.
- Geographic imbalance: Overrepresentation from India.
- Program concentration: Applications clustered in specific programs.(Computer science, business analyst)
- Admission trends: Notable spike in 2023



STRATEGIC RECOMMENDATIONS

1. Prioritise high-performing programs and regions while expanding into new markets.
2. Shift focus from outreach volume to outreach effectiveness.
3. Strengthen follow-up and mid-funnel engagement processes.
4. Balance counselor workload to ensure consistent applicant experience.
5. Leverage historical trend peaks to plan future outreach timing.

CONCLUSION

This project strengthened both our technical and analytical capabilities, as we learned to derive insights from incomplete and noisy datasets.
By transforming messy CRM data into a strategic recruitment asset, our team improved DePaul's recruitment ROI and outreach performance.